

Dondapati Karthikeya

Software Engineer – Python / Full Stack Development

Hyderabad, Telangana — +91-6309554591 — karthikeyadondapati94@gmail.com — LinkedIn — GitHub

Career Objective

B.Tech (CSE – AI & ML) graduate with a solid foundation in Python, Data Structures, OOP, and full-stack development using Django, Flask, FastAPI, and React.js. Hands-on experience building REST APIs, working with SQL and MongoDB databases, and using Git through academic projects and a software development internship. Seeking an entry-level Software Engineer role to apply strong problem-solving skills to real-world systems.

Education

B.Tech in Computer Science Engineering (AI & ML)	2022 – 2026
Mallareddy University, Hyderabad	Percentage: 83%
Intermediate Education (MPC)	2020 – 2022
Sri Vidya Junior College, Kodad	Percentage: 94.5%
Secondary School Education (SSC)	2019 – 2020
Sri Chaitanya Techno School, Kodad	Percentage: 95%

Technical Skills

- **Programming Languages:** Python, SQL
- **Core CS:** Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks
- **Full Stack Technologies:** Django, Flask, FastAPI, REST API, JWT Authentication, React.js, HTML, CSS
- **Databases:** MySQL, PostgreSQL, MongoDB, Vector Databases
- **AI & ML:** Machine Learning, Scikit-learn, Large Language Models, RAG, Computer Vision
- **Tools & Cloud:** Git, GitHub, Postman, AWS Cloud Foundations, CI/CD Basics
- **Professional Skills:** Problem Solving, Analytical Thinking, Communication, Team Collaboration, Adaptability

Projects

AI-Powered Multi-Agent India Travel Planner

Technology Stack: Python, Next.js, Groq API, Vercel AI SDK, PostgreSQL (Supabase), REST API, JWT

- Built a backend-driven **multi-agent GenAI workflow** with 3 chained LLM agents (Destination Analyzer, Route Optimizer, Day Planner) using Groq's Llama-3.3 to orchestrate end-to-end itinerary generation via **REST APIs**.
- Designed sequential agent-to-agent prompting and tool-use logic to analyze destinations, optimize routes, and produce structured, dynamic day-wise plans.
- Integrated **Supabase** for auth and storage, and **jsPDF** for downloadable itinerary PDFs.

Social Bot Detection System

Technology Stack: Python, Django, MySQL, Scikit-learn, XGBoost, TensorFlow/Keras

- Built a full-stack **Django** web application with role-based access control (Admin, Model Builder, User) for detecting social media bots from behavioral account data.
- Trained and compared **5 ML models** (Logistic Regression, Random Forest, SVM, **XGBoost**, and a **Keras ANN**) on engineered behavioral features, selecting the best-performing model for deployment.
- Handled class imbalance using resampling techniques and feature scaling (StandardScaler) to improve model reliability.
- Designed a **MySQL**-backed authentication system and a real-time prediction pipeline with confidence-based classification.

Liver Disease Prediction System

Technology Stack: Python, Flask, Scikit-learn, SQL

- Cleaned and analyzed a real-world medical dataset; built a **Scikit-learn** machine learning model to predict liver disease risk.
- Improved model performance by **12%** through feature engineering and hyperparameter tuning.
- Built and deployed a **Flask** web application enabling users to input health data and receive real-time predictions.

Internship

Python Developer Intern — DRDOT Solutions

May 2025 – Sep 2025

- Built a **Number Plate Recognition system** in Python using **YOLOv5** and OCR, reaching around **95% accuracy** after preprocessing and model tuning.
- Developed a **React-based restaurant ordering platform** with an **LLM-powered chatbot** that used **JSON** menu data to recommend dishes and let customers place orders conversationally.
- Debugged and tested application code to fix issues and improve overall performance and reliability.

Certifications

- AWS Academy Cloud Foundations
- Deloitte Data Analytics Virtual Experience
- Accenture Data Analytics Virtual Experience